



Dwight Look College of
ENGINEERING
TEXAS A&M UNIVERSITY

Team 45: LES Parts Database UI Final Presentation

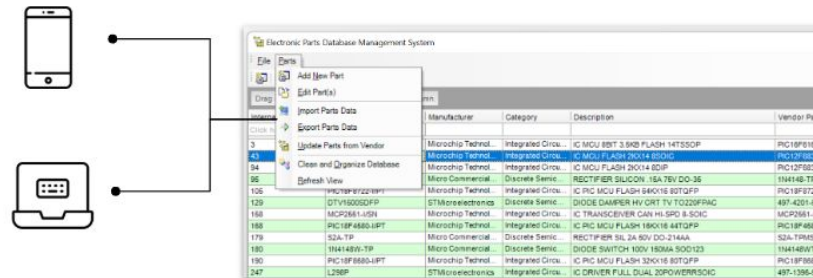
Anthony Beneventi, Jamaal Bloom, Jaswin Jabbal

Sponsor: John Lusher

TA: Sabyasachi Gupta

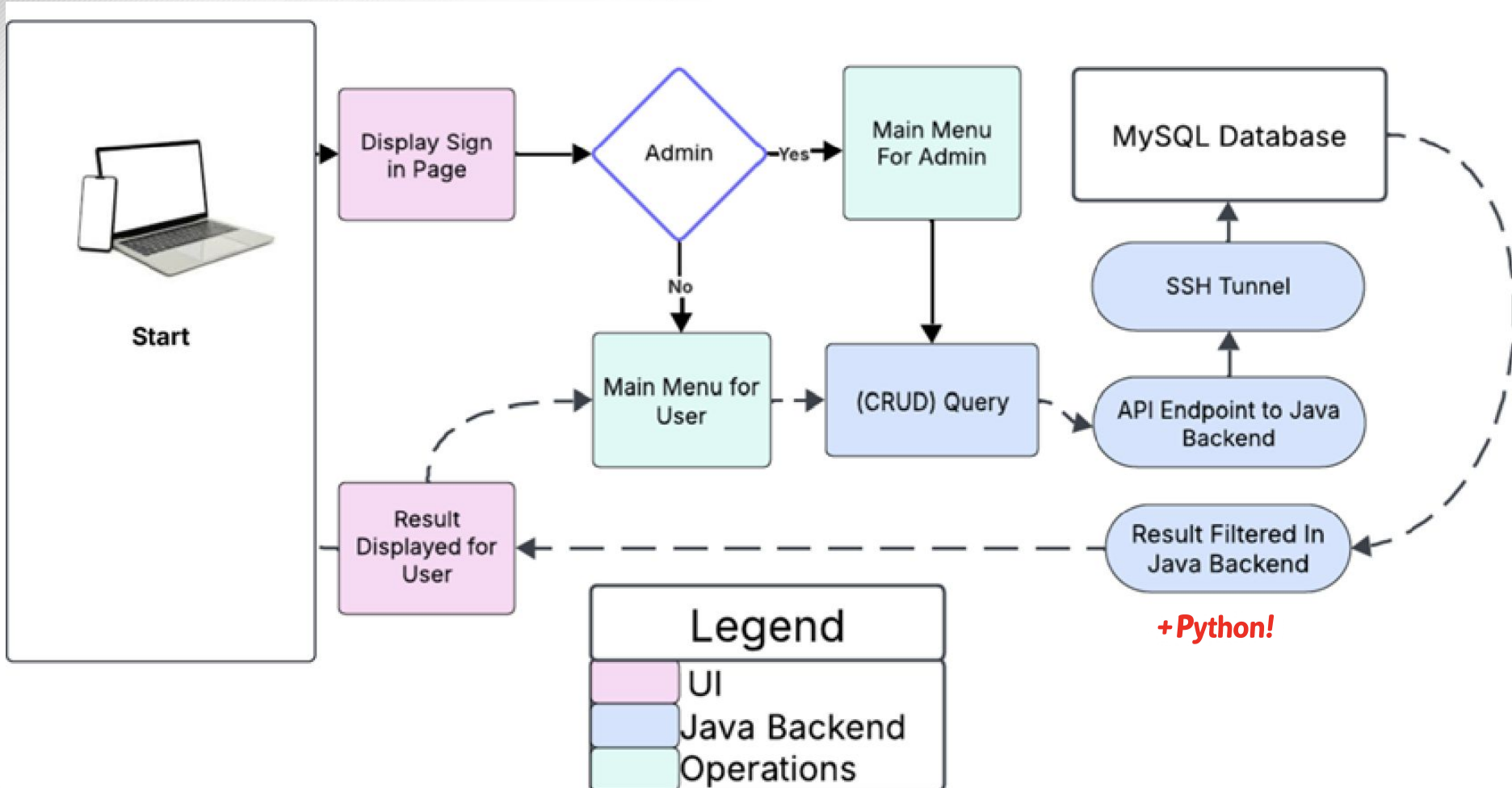
Problem Overview

LES handles **electronic part management** via MySQL Workbench, relying on **manual data entry** and lacks multi-user access, hindering productivity and **scalability** for database operations.



Our team has developed full-stack applications for both **mobile** and **desktop**, streamlining operations by allowing **concurrent users** to work on the same database through **automated data population (APIs)** and **intelligent filtering** of part information, enabling real-time collaboration and efficient data management.

Integrated System Diagram



Web Application UI

Jaswin Jabbal



LES_Parts_Catalog/	
__pycache__/	
sql/	
create_users_table.sql	# SQL schema for user/admin table
static/	
LES.jpg	# Logo used in frontend
templates/	# HTML templates/forms for Flask
add_part.html	
delete.html	
digikey_add.html	
filter.html	
index.html	
layout.html	# Base UI template extended to all other pages
login.html	
profile.html	
register.html	
sort.html	
tag_part.html	
update.html	
utils/	# Backend logic and helpers
__pycache__/	
models.py	# SQLAlchemy models ensure scalable DB operations
query_helpers_api.py	# Backend integration for all CRUD functions + Digikey API
venv/	# Virtual environment ensures correct interpreter/packages
.env	# Environment variables ensure secure settings
app.py	# Flask app entry point to ensure correct routing of pages
README.md	# Clear project overview and setup documentation
requirements.txt	# Python dependencies ensure consistent variables across machines
run_app.bat	# Windows launch script ensures hassle-free setup!



Web Application UI

Challenges

- Dynamic HTML rendering for complex data.
- SSH tunnel integration.
- Lack of support for concurrent users.
- Handling Digikey API interactions.

Solutions

- Using Jinja2 templating via layout and modular route logic.
- Scripting automatic tunnel start when the application is built + run.
- Developing login/auth/profile systems with Flask session management.
- Creating helper module for API calls + input validation .



Dwight Look College of

ENGINEERING
TEXAS A&M UNIVERSITY

Web Application UI

[LES Parts Database UI - Final Presentation](#)



Python Backend

Engineering Design Accomplishments

Jamaal Bloom

Functions:

- Refresh Token
- Use access token
- Filter (Web)
- Sort (Web)
- Tag (add and delete)
- Add (Digikey and Mouser)
- Update
- Delete (Web)

Challenges and solutions:

- Digikey and mouser APIs
- SSH Tunnel

Integration design:

- Vercel API (Connect through github)
- SSH Tunnel Connection

Python Backend

Tested/Method:

- Functions/ test by using arbitrary values
- Supplier APIs/ referenced workbench and supplier website
- Database Sync/ called function on different instances of mobile and web app.

Allied Ele...	845-0101	Fuses	No	Potentially Obsolete Due to being unable to find...	0	0	15.89	12.72	12.38	NULL
Digi-Key	0034.1741-ND	Circuit Protection	No	Potentially Obsolete Due to being unable to find...	1	0	1.668	1.235	1.235	NULL
Digi-Key	283-2841-ND	Circuit Protection	Yes	Got	1	0	2.46	1.4387	1.09805	NULL
Digi-Key	1703034-ND	Connectors, Interconnects	No	Potentially Obsolete Due to being unable to find...	1	0	3.37	3.37	3.37	NULL



BK/GMA-3-R

DigiKey Part Number: 283-2841-ND

Manufacturer: Eaton - Electronics Division

Manufacturer Product Number: BK/GMA-3-R

Description: FUSE GLASS 3A 250VAC 5X20MM

Manufacturer Standard Lead Time: 12 Weeks

Customer Reference:

Detailed Description: 3 A 250 V AC DC Fuse Cartridge, Glass Requires Holder 5mm x 20mm

Dataset: [Dataset](#)

In-Stock: 544

Can ship immediately

QUANTITY:

[Add to List](#) [Add to Cart](#)

All prices are in USD

Bulk

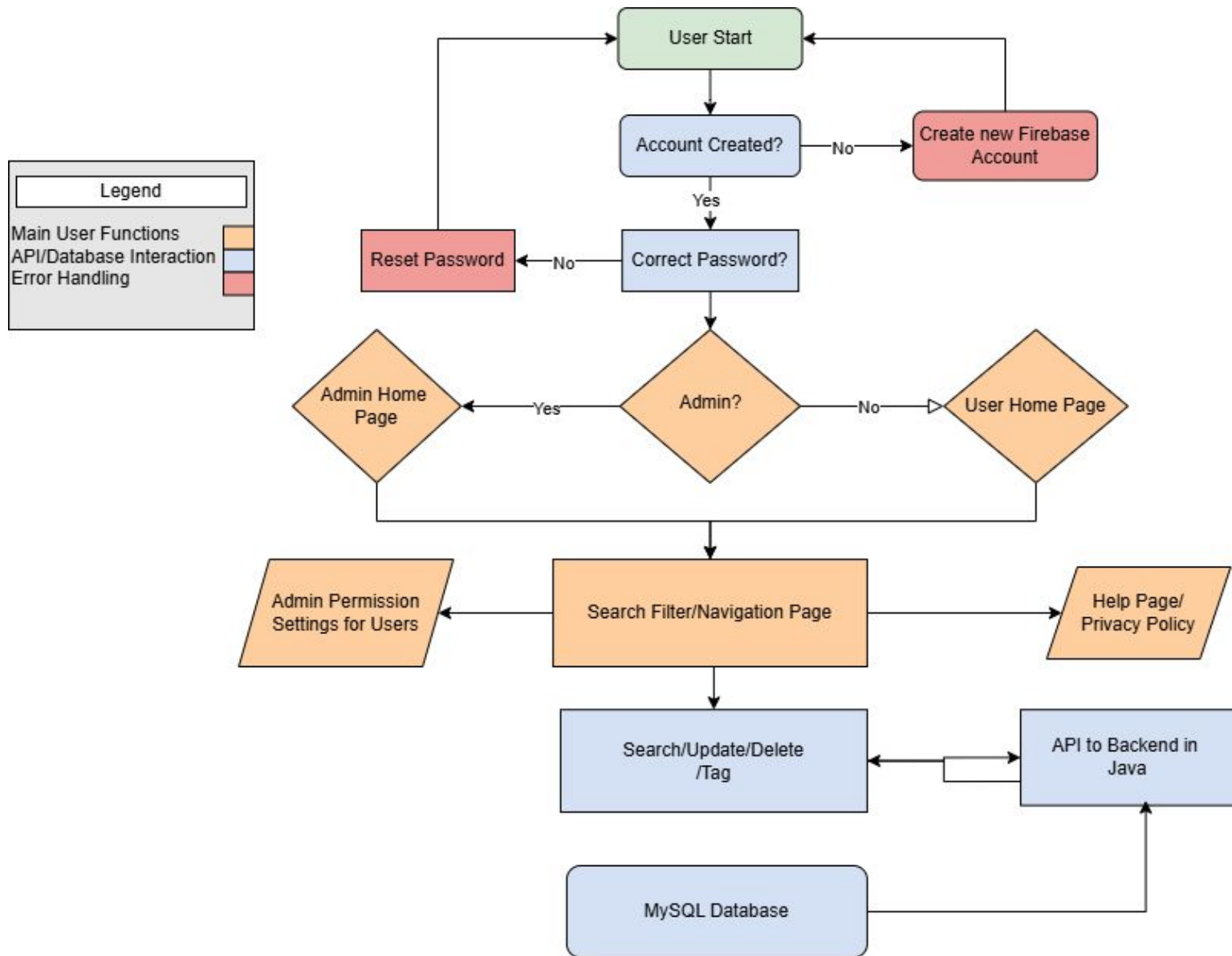
QUANTITY	UNIT PRICE	EXT PRICE
1	\$2.46000	\$2.46
5	\$2.10900	\$10.55
10	\$1.80900	\$18.09
25	\$1.55940	\$38.99
100	\$1.43870	\$143.87
300	\$1.26537	\$379.61
500	\$1.19168	\$595.84
1,000	\$1.09805	\$1,098.05

Product Attributes

TYPE	DESCRIPTION	SELECT ALL
Category	Circuit Protection	☒
	Fuses	☒
Mfr	Eaton - Electronics Division	☐
Series	GMA	☐

IOS Application Flowchart

Anthony Beneventi





IOS Challenges & Solutions

Anthony Beneventi

Challenges

- Using Java code in order to deploy an IOS App.
- SSH Tunnel Integration Errors.
- User interface display error where user could not view results.
- Handling unexpected inputs from user in App Queries.

Solutions

- Using Flutterflow as a proxy in order to bridge the Android IOS gap.
- Connecting and disconnecting the SSH manually in order to decrease system runtime.
- Introduced reload and refresh functions into every major operation.
- Backend Error handling to convert incorrect data types and proper Error messages given back to the User.



IOS Subsystem Results

Anthony Beneventi

<https://www.youtube.com/watch?v=jERSYX5G1HI>

Link to previous demo with further function detail:
<https://youtu.be/0rWJT-cg5A4>

Item	Met	Runtime
Run API/Query through multiple devices simultaneously	Pass	~0.2-0.3 seconds
Allow Admin User to permissions	Pass	~0.2-1 seconds
Error handle incorrect user input	Pass	~0.2-1 seconds
App can run Jamaal's Update & Tag Functions.	Pass	~1-10 seconds
Average User-Interface load time	Pass	~2 seconds



Conclusions

System Changes:

- Originally developed for a cloud based database but changed to ssh connection.

Current status on deliverables:

- Integration completed between subsystems and the original MySQL database.
- Error handling and error messages for user convenience.
- Multiple user functionality confirmed and validated.
- Deployment of application complete.



Dwight Look College of

ENGINEERING
TEXAS A&M UNIVERSITY

Team 45: Anthony Beneventi, Jamaal Bloom, and Jaswin Jabbal

Thank you! Questions?